Grizon AI — Master Technical Documentation

Platform Audit + Legal Agent Architecture + Implementation Roadmap

Prepared by: Senior AI Systems Architect (20+ Years) **Date:** August 7, 2026 **Platform Version:** Grizon v0.4.0 **Scope:** Full platform audit (live session) + Legal Research Agent deep analysis + Priority roadmap

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> [!IMPORTANT] This document is a comprehensive technical audit prepared for the Grizon AI founding team. It combines a live session analysis of `app.grizonai.com`, review of all architecture documentation, and a deep analysis of the Legal Research & Case Analysis Agent system prompt. All findings are based on observed behavior — not assumptions.

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1. EXECUTIVE SUMMARY

Grizon AI is a **next-generation AI agent workspace platform** (v0.4.0) with a functioning core — authentication, streaming chat, PDF document analysis, credit billing via PhonePe UPI, and a well-designed multi-LLM routing engine called **Project Brain**.

However, the platform has **4 critical gaps** that must be resolved before serious growth:

#	Gap	Business Impact
1	**Bot doesn't know its own pricing** (RAG broken)	Every prospect who asks gets wrong answer — sales killer
2	**Composer input disappears** after any response	Core UX broken — users can't have multi-turn conversations
3	**Zero hallucination detection** active in production	Platform's core promise (zero hallucination) is not delivered
4	**Legal Agent tools incomplete** (file_gen, Hindi OCR)	Legal product cannot launch without these

Overall Platform Health: 5.5 / 10 Legal Agent Design Quality: 7 / 10 (excellent concept, tool gaps to resolve)

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2. CURRENT SYSTEM ARCHITECTURE

2.1 Live Confirmed Architecture

```
graph TB
    subgraph CLIENT["CLIENT LAYER"]
        A["Next.js Frontend<br/>(React + Tailwind CSS)<br
    end

    subgraph AUTH["AUTHENTICATION"]
        B["Email-first Auth<br/>Email → Password → /chat"
        C["Auth Guard<br/>Server-side redirect"]
    end

    subgraph GATEWAY["API GATEWAY"]
        D["Node.js / Express<br/>API Gateway"]
    end

    subgraph CACHE["CACHE LAYER"]
        E["Redis<br/>Semantic Cache<br/>cosine ≥ 0.92<br/
    end

    subgraph VECTOR["VECTOR DATABASE"]
        F["Pinecone<br/>grizonai-knowledge<br/>category:
    end

    subgraph DB["DATABASE"]
        G["PostgreSQL<br/>users · wallets · conversations
    end

    subgraph PAYMENT["PAYMENT"]
        H["PhonePe UPI AutoPay<br/>SHA256 X-VERIFY<br/>✓
    end

    subgraph BRAIN["PROJECT BRAIN — LLM ROUTER"]
        I["Claude 3.5 Sonnet<br/>PRIMARY"]
        J["DeepSeek V3/R1<br/>FALLBACK"]
        K["GPT-4o/5<br/>TERTIARY"]
    end

    subgraph OCR["DOCUMENT PIPELINE"]
```

```
    L["PyMuPDF<br/>Digital PDFs ✓"]
    M["PaddleOCR<br/>Mixed Hindi+English"]
    N["Sarvam Vision<br/>Hindi Handwritten ✗ Not wir
end

subgraph SARVAM["SARVAM AI LAYER — NOT YET INTEGRATED
    O["Saaras v3 STT<br/>Voice → Text ✗"]
    P["Bulbul v2 TTS<br/>Text → Audio ✗"]
    Q["sarvam-translate<br/>22 Indian languages ✗"]
    R["text-lid<br/>Language Detection ✗"]
end

A --> B --> C --> D
D --> E
D --> F
D --> G
D --> H
D --> BRAIN
I -->|"fail 429/503"| J -->|"fail"| K
D --> OCR

style F fill:#ff4444,color:#fff
style O fill:#ff9944,color:#fff
style P fill:#ff9944,color:#fff
style Q fill:#ff9944,color:#fff
style R fill:#ff9944,color:#fff
style N fill:#ff9944,color:#fff
```

2.2 What's Confirmed Working (Live Session)

Component	Status	Measured Performance
User Authentication	✓ Working	Email → password, session persisted
Auth Guard	✓ Working	/chat → login redirect when logged out

Text Chat (streaming)	✓ Working	1.4s – 3.3s TTFT
PDF Document Upload	✓ Working	5.7s, 661KB PDF, ZERO hallucination on doc content
Credit Deduction	✓ Working	3 credits/text query, 6 credits/PDF query
Billing Dashboard	✓ Working	Pro ₹1,999/month, renews Aug 31 2026
PhonePe UPI AutoPay	✓ Working	Mandate linked, recurring active
Model Selector	✓ Working	Auto mode + General Assistant dropdown
Sidebar Navigation	✓ Working	New chat, search, folders, agents, settings
Rate Limiting	✓ Working	60/hr · 400/day · 2000/week · 10000/month
Mobile Collapse	✓ Working	Hamburger drawer at mobile width
Streaming Responses	✓ Working	Token-by-token streaming

3. DATA FLOW DIAGRAMS

3.1 Complete Message Processing Flow (Current)

```
sequenceDiagram
    participant U as User (Browser)
    participant FE as Next.js Frontend
    participant GW as API Gateway
    participant RC as Redis Cache
```

```
    participant PC as Pinecone RAG
    participant PB as Project Brain
    participant DB as PostgreSQL

    U->>FE: Types message + sends
    FE->>GW: POST /api/chat { message, conversationId }
    GW->>DB: Check user credits (SELECT FOR UPDATE)
    DB-->>GW: Credits available

    Note over GW,RC: ⚠ Cache status unknown
    GW->>RC: Semantic similarity lookup (cosine ≥ 0.92)

    alt Cache HIT
        RC-->>GW: Cached response
        GW-->>FE: Return cached answer (fast)
    else Cache MISS
        Note over GW,PC: 🔴 BROKEN — not being queried
        GW->>PC: Query grizonai-knowledge (company_docs)
        PC-->>GW: Retrieved context chunks

        GW->>PB: Send message + context to Claude 3.5 Son

        alt Claude succeeds
            PB-->>GW: English response (streaming)
        else Claude fails (429/503)
            PB->>PB: Fallback to DeepSeek V3
            PB-->>GW: Response from DeepSeek
        end

        GW->>DB: Save message, deduct credits
        GW-->>FE: Stream response via SSE
    end

    FE-->>U: Render response + show metadata
    Note over FE,U: Shows: model | time | credits used
    Note over FE,U: ✖ MISSING: confidence badge, groundi
```

3.2 Document/OCR Processing Flow

```
flowchart TD
    A["User uploads document"] --> B{"File type?"}

    B -->|"Digital PDF"| C["PyMuPDF direct extraction<br/
    B -->|"Scanned image/PDF"| D{"Language?"}
    B -->|"Image file"| D

    D -->|"Hindi / Devanagari"| E{"Handwritten?"}
    D -->|"English printed"| F["Surya OCR / Tesseract<br/
    D -->|"Mixed Hindi+English"| G["PaddleOCR<br/>⚠  Stat

    E -->|"YES"| H["Sarvam Vision API<br/>🔴 NOT WIRED<br
    E -->|"NO"| I["Sarvam Vision API<br/>🔴 NOT WIRED<br/

    C --> J["OCR Post-Processing<br/>Unicode normalize ·
    F --> J
    G --> J
    H --> J
    I --> J

    J --> K["Prompt Preprocessing<br/>Intent classify · E

    K --> L["Project Brain<br/>temp=0.1 for documents<br/

    L --> M["Response Verifier<br/>🔴 NOT WIRED IN PRODUC

    M -->|"confidence ≥ 0.7"| N["✓ Send response to user
    M -->|"confidence < 0.7"| O["⚠  Add LOW CONFIDENCE wa

    style H fill:#ff4444,color:#fff
    style I fill:#ff4444,color:#fff
    style M fill:#ff4444,color:#fff
```

4. SYSTEM HEALTH SCORECARD

```
xychart-beta
    title "Grizon AI v0.4.0 — Component Health Scores (ou
    x-axis ["Auth", "Chat", "OCR/Docs", "Hallucination",
    y-axis "Score" 0 --> 10
    bar [9, 7, 8, 3, 1, 6, 9, 3, 0, 2]
```

Category	Score	Grade	Notes
Authentication	9/10	**A**	Solid email auth, server-side guard
Chat Functionality	7/10	**B**	Streaming works; composer bug critical
Document OCR (digital PDF)	8/10	**B+**	PyMuPDF excellent; Hindi OCR missing
Hallucination Prevention	3/10	**D**	Architecture designed, NOT deployed
Company Knowledge RAG	1/10	**F**	Pinecone broken — bot ignores own pricing
Mobile Experience	6/10	**C+**	Collapse works; iOS zoom risk present
Payment System	9/10	**A**	PhonePe AutoPay fully working
Voice Input	3/10	**D**	UI stub only — no Sarvam STT wired
Multilingual Support	0/10	**N/A**	English only; Sarvam not integrated
Response Verification	2/10	**F**	ResponseVerifier built but not active
OVERALL	**5.5/10**	**C+**	Strong foundation, critical gaps remain

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5. CRITICAL BUGS — PLATFORM CORE

BUG #1 — Bot Doesn't Know Its Own Pricing

Severity: 🔴 CRITICAL | Impact: Sales-killer

```
graph LR
    A["User asks:<br/>'What are your pricing plans?'"] --
    B --> C{"queryPinecone called?"}
    C -->|"NO — BUG"| D["Project Brain gets<br/>ZERO comp
    D --> E["Claude responds:<br/>'I don't know any prici
    E --> F["🔴 WRONG ANSWER<br/>User is ON the Pro plan!

    C -->|"SHOULD DO THIS"| G["Pinecone grizonai-knowledg
    G --> H["Retrieved: Plans, Pricing,<br/>Features, FAQ
    H --> I["Project Brain gets full context"]
    I --> J["✅ Correct answer with exact prices"]

    style F fill:#ff4444,color:#fff
    style J fill:#00aa44,color:#fff
    style C fill:#ff9900,color:#fff
```

Root Cause: `chatController.js` is not calling `queryPinecone()` for general chat queries. The Pinecone index (`grizonai-knowledge`) is also likely empty of company knowledge documents.

What needs to be populated in Pinecone:

- Pricing plans: Free (₹0), Basic (₹499), Pro (₹1,999), Elite (₹4,999)
- Credits per plan: 500 / 2,000 / 6,000
- Features per tier
- Credit costs per query type (3 credits text / 6 credits PDF)
- Company FAQ, mission, contact

BUG #2 — Composer Input Disappears After Long Response

Severity: 🔴 CRITICAL | Impact: Core UX broken for all users

```
graph TD
    A["User sends message"] --> B["Bot gives long respons
    B --> C["Page-level scroll activates<br/>(BUG: wrong
    C --> D["Chat messages push composer<br/>from Y=945px
    D --> E["🟡 Composer INPUT BOX<br/>is now OFF-SCREEN"
    E --> F["User cannot type<br/>follow-up message"]

    G["CORRECT BEHAVIOR"] --> H["Messages container scrol
    H --> I["Composer ALWAYS pinned<br/>flex-shrink: 0 at
    I --> J["✓ Input always visible<br/>regardless of me

    style E fill:#ff4444,color:#fff
    style J fill:#00aa44,color:#fff
```

BUG #3 — Voice Input Textarea Hidden After Recording Stops

Severity: 🟡 MEDIUM | Impact: Confusing UX on voice flow

After stopping voice recording, the textarea doesn't regain focus/ visibility. User must click empty space to recover.

Fix: Add `textareaRef.current?.focus()` with 150ms delay after recording stop.

BUG #4 — Response Verifier / LLM-as-Judge NOT Active

Severity: 🟡 HIGH | Impact: Platform's core anti-hallucination promise not delivered

The `ResponseVerifier` class (LLM-as-Judge, number-check, groundedness scoring) is fully designed in architecture docs but **not wired** into the production chat pipeline. Result: the bot confidently stated wrong pricing information with zero warning or confidence indicator.

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6. LEGAL RESEARCH AGENT ARCHITECTURE

6.1 Agent Overview

The **Legal Research & Case Analysis Agent** is a specialized AI persona layered on top of Project Brain. It transforms Grizon AI from a general chatbot into a structured multi-step legal reasoning engine, purpose-built for **Indian law**.

```
graph TB
    subgraph INPUT["USER INPUT"]
        A1["Text query<br/>(legal question)"]
        A2["Voice query<br/>(Hindi/English)"]
        A3["Document upload<br/>(FIR, contract, judgment)
        A4["Image upload<br/>(scanned court notice)"]
    end

    subgraph PREPROCESS["PRE-PROCESSING"]
        B1["Language Detection<br/>text-lid (Sarvam)"]
        B2["STT Transcription<br/>Saaras v3 (Sarvam)"]
        B3["Transliterate<br/>Romanized → Devanagari"]
        B4["Translate to English<br/>sarvam-translate"]
        B5["Document / Image Analysis<br/>document_analys
    end

    subgraph WORKFLOW["6-STAGE LEGAL REASONING WORKFLOW"]
        S1["Stage 1: Legal Issue Identification<br/>Domai
        S2["Stage 2: Statutory Lens<br/>Retrieve Acts · S
        S3["Stage 3: Case Law & Precedent<br/>SC + HC + T
        S4["Stage 4: Argument Analysis<br/>FOR: 3-4 point
        S5["Stage 5: Preliminary Assessment<br/>Strong /
        S6["Stage 6: Procedural & Risk Check<br/>Forum ·
    end
```

```
    subgraph OUTPUT["DUAL OUTPUT"]
        O1["Part A: Chat Summary<br/>1 line: issue + Act/
        O2["Part B: Full Report Document<br/>file_gen / d
    end

    A1 & A2 --> B1 & B2 & B3 & B4
    A3 --> B5
    A4 --> B5
    B1 & B2 & B3 & B4 & B5 --> S1
    S1 --> S2 --> S3 --> S4 --> S5 --> S6
    S6 --> O1
    S6 --> O2

    style S3 fill:#ff9900,color:#000
    style O2 fill:#ff9900,color:#000
```

6.2 The 6-Stage Workflow — Detailed Flowchart

```
flowchart TD
    START(["User submits legal query"]) --> S1

    subgraph S1["STAGE 1: ISSUE IDENTIFICATION"]
        S1A["Identify legal domain<br/>Criminal / Civil /
        S1B{"Facts clear?"}
        S1C["Ask clarifying questions<br/>before proceedi
        S1A --> S1B
        S1B -->|"No"| S1C --> S1A
        S1B -->|"Yes"| S1D["Identify jurisdiction<br/>Cen
    end

    S1D --> S2

    subgraph S2["STAGE 2: STATUTORY LENS"]
        S2A["Retrieve applicable Acts<br/>Sections · Arti
        S2B["Check amendment status<br/>IPC → BNS (post J
        S2C["Summarize scope<br/>Purpose · Exceptions · P
        S2A --> S2B --> S2C
    end
```

```
S2C --> S3

subgraph S3["STAGE 3: CASE LAW & PRECEDENT"]
    S3A["Search SC judgments<br/>Binding on all court
    S3B["Search HC judgments<br/>Binding in that stat
    S3C["Search Tribunal decisions<br/>NCLT · Labour
    S3D["If document uploaded<br/>document_analyse /
    S3E["Compile citations [1][2][3]<br/>Summarize pr
    S3A & S3B & S3C & S3D --> S3E
end

S3E --> S4

subgraph S4["STAGE 4: ARGUMENT ANALYSIS"]
    S4A["FOR the user<br/>3-4 supporting legal points
    S4B["AGAINST the user<br/>3-4 counterarguments"]
    S4C["Weigh objectively<br/>No favouring either si
    S4A & S4B --> S4C
end

S4C --> S5

subgraph S5["STAGE 5: PRELIMINARY ASSESSMENT"]
    S5A{"Legal position strength?"}
    S5B["STRONG — High Confidence"]
    S5C["MODERATE — Medium Confidence"]
    S5D["WEAK — Low Confidence"]
    S5E["State the ONE pivotal fact<br/>that could ch
    S5A --> S5B & S5C & S5D
    S5B & S5C & S5D --> S5E
end

S5E --> S6

subgraph S6["STAGE 6: PROCEDURAL & RISK CHECK"]
    S6A["Identify correct forum<br/>SC / HC / Distric
    S6B["Limitation period check<br/>⚠  MOST COMMON F
    S6C["Required documents checklist"]
    S6D["Procedural risks<br/>Evidentiary gaps · Comp
```

```
        S6A --> S6B --> S6C --> S6D
    end

    S6D --> OUT

    subgraph OUT["DUAL OUTPUT"]
        OA["Part A: Chat Summary<br/>Issue · Assessment ·
        OB["Part B: Full Report<br/>via file_gen / docume
        S6D --> OA
        S6D --> OB
    end

    style S3 fill:#fff3cd,color:#000
    style S5B fill:#d4edda,color:#000
    style S5C fill:#fff3cd,color:#000
    style S5D fill:#f8d7da,color:#000
    style S6B fill:#f8d7da,color:#000
```

6.3 Legal Agent Tool Requirements vs. Current Status

Tool	Purpose	Current Status	Risk if Missing
`document_analyse`	Read FIRs, contracts, judgments, PDFs	✓ Partially working (digital PDF)	Medium — scanned docs fail
`web_search`	Retrieve statutes, case law, notifications	✓ Present	High — must reach verified sources
`image_analyse`	Read scanned court notices, handwritten docs	✗ Sarvam Vision not wired	HIGH — most Indian legal docs are scanned

`file_gen` / `documentCreation`	Generate Part B full report document	✖ Not confirmed working	CRITICAL — entire Part B output fails

7. LEGAL AGENT RISK ANALYSIS

7.1 Risk Heatmap

```
quadrantChart
    title Legal Agent Risks — Likelihood vs Impact
    x-axis "Low Likelihood" --> "High Likelihood"
    y-axis "Low Impact" --> "High Impact"
    quadrant-1 "CRITICAL — Fix Before Launch"
    quadrant-2 "Monitor Closely"
    quadrant-3 "Accept"
    quadrant-4 "Watch"
    "Legal Citation Hallucination": [0.8, 0.9]
    "IPC→BNS Section Confusion": [0.7, 0.85]
    "file_gen Failure": [0.75, 0.7]
    "Stale Legal Information": [0.6, 0.75]
    "Jurisdiction Ambiguity": [0.5, 0.8]
    "Disclaimer Missing": [0.3, 0.9]
    "Hindi Doc OCR Failure": [0.7, 0.6]
    "Rate Limit on Legal Queries": [0.4, 0.4]
```

7.2 The IPC → BNS Transition — Critical Context

India's criminal law underwent a complete overhaul effective **July 1, 2024**:

```
graph LR
    subgraph OLD["OLD LAW — REPEALED"]
        A["IPC 1860<br/>Indian Penal Code"]
```

```
    B["CrPC 1973<br/>Criminal Procedure Code"]
    C["Indian Evidence Act 1872"]
end

subgraph NEW["NEW LAW — IN FORCE FROM JULY 1, 2024"]
    D["BNS 2023<br/>Bharatiya Nyaya Sanhita"]
    E["BNSS 2023<br/>Bharatiya Nagarik Suraksha Sanhi
    F["BSA 2023<br/>Bharatiya Sakshya Adhiniyam"]
end

A -->|"Replaced by"| D
B -->|"Replaced by"| E
C -->|"Replaced by"| F

subgraph MAPPING["SECTION NUMBER CHANGES"]
    G["IPC 302 Murder → BNS 101"]
    H["IPC 420 Cheating → BNS 316"]
    I["IPC 498A Cruelty → BNS 84"]
    J["IPC 376 Rape → BNS 63"]
end

style A fill:#ff9999
style B fill:#ff9999
style C fill:#ff9999
style D fill:#99ff99
style E fill:#99ff99
style F fill:#99ff99
```

> [!CAUTION] If the Legal Agent cites IPC Section 302 for an offence that occurred **after July 1, 2024**, that is **legally wrong**. The agent must detect the date of offence and apply the correct law. This is the #1 legal AI error risk in India right now.

7.3 Indian Court Hierarchy & Limitation Periods

```
graph TD
    SC["🏛  SUPREME COURT OF INDIA<br/>Binding on ALL cou
    HC["⚖  HIGH COURTS x25<br/>Binding in their state onl
```

```
    DC["🏢 DISTRICT & SESSIONS COURTS<br/>Civil suits · C
    FC["🧒👩👧 FAMILY COURTS<br/>Matrimonial · Maintenanc
    LC["⛑️ LABOUR COURTS / INDUSTRIAL TRIBUNAL<br/>Indust
    CF["  CONSUMER FORUMS<br/>District / State / National
    NCLT["🏦 NCLT / NCLAT<br/>Company law · Insolvency<br
    ITT["💰 INCOME TAX TRIBUNAL<br/>Tax disputes<br/>Limit

    SC --> HC --> DC & FC & LC & CF & NCLT & ITT

    style SC fill:#1a1a2e,color:#fff
    style HC fill:#16213e,color:#fff
```

---

8. FEATURE IMPLEMENTATION STATUS

```
graph LR
    subgraph DONE["✓ WORKING IN PRODUCTION"]
        D1["Email Authentication"]
        D2["Auth Guard (server-side)"]
        D3["Text Chat + Streaming"]
        D4["Digital PDF Extraction"]
        D5["Credit Deduction System"]
        D6["PhonePe UPI AutoPay"]
        D7["Billing Dashboard"]
        D8["Model Selector (Auto)"]
        D9["Sidebar Navigation"]
        D10["Rate Limiting"]
        D11["Mobile Collapse Layout"]
    end

    subgraph BROKEN["🔴 BROKEN / CRITICAL"]
        B1["Pinecone RAG — company docs<br/>Bot doesn't k
        B2["Composer Push Bug<br/>Input disappears after
        B3["Response Verifier<br/>LLM-as-Judge not wired"
    end

    subgraph STUB["🟡 STUB / PARTIAL"]
```

```
        S1["Voice Input<br/>UI exists, no Sarvam STT"]
        S2["Scanned Document OCR<br/>Hindi/handwritten pa
        S3["Redis Semantic Cache<br/>Status unknown"]
        S4["Confidence Badges<br/>API returns data, UI do
    end

    subgraph MISSING["❌ NOT YET BUILT"]
        M1["Sarvam Saaras v3 STT"]
        M2["Sarvam Bulbul v2 TTS"]
        M3["Hindi/Multilingual support"]
        M4["file_gen document creation"]
        M5["Indian Kanoon API integration"]
        M6["Legal RAG (BNS/BNSS/BSA/Constitution)"]
        M7["PhonePe voice checkout"]
        M8["WhatsApp integration"]
        M9["OCR quality indicator UI"]
        M10["IPC→BNS transition detection"]
    end
```

---

9. PRIORITY IMPLEMENTATION ROADMAP

```
gantt
    title Grizon AI — Implementation Roadmap (Aug-Sep 202
    dateFormat  YYYY-MM-DD
    section WEEK 1 — Stop the Bleeding
    Fix Composer Push Bug (CSS flex)       :crit, done, w
    Populate Pinecone company docs          :crit, w1b, 2
    Wire queryPinecone to chatController    :crit, w1c, a
    Fix Voice Textarea Focus Bug            :w1d, 2026-08
    Fix iOS textarea font-size (16px)       :w1e, 2026-08

    section WEEK 2 — Hallucination Prevention
    Wire ResponseVerifier to chat flow      :crit, w2a, 2
    Add confidence badges to frontend       :w2b, after w
    Task-adaptive temperature (doc vs chat) :w2c, 2026-08
    Add "which model" to response metadata  :w2d, 2026-08
```

```
    Fix two credit pool UX confusion        :w2e, 2026-08

    section WEEK 3 — Legal Agent Foundation
    Integrate Indian Kanoon API             :crit, w3a, 2
    Build legal RAG (BNS, BNSS, BSA, Const):crit, w3b, 20
    Implement file_gen tool                 :w3c, 2026-08
    Hard-code legal disclaimer footer       :w3d, 2026-08
    IPC→BNS date detection logic            :w3e, 2026-08

    section WEEK 4 — Sarvam AI Integration
    Wire Saaras v3 STT to voice button      :w4a, 2026-08
    Wire Bulbul v2 TTS audio output         :w4b, 2026-08
    Language detection (text-lid) per query :w4c, 2026-08
    Translate query to English for RAG      :w4d, 2026-09
    Translate responses back to user lang   :w4e, 2026-09
    Wire Sarvam Vision OCR for Hindi docs   :w4f, 2026-09
```

Priority Matrix

```
quadrantChart
    title Implementation Priority Matrix
    x-axis "Low Effort" --> "High Effort"
    y-axis "Low Business Impact" --> "High Business Impac
    quadrant-1 "Do Later (Plan)"
    quadrant-2 "Quick Wins — Do First"
    quadrant-3 "Deprioritize"
    quadrant-4 "Strategic — Plan Carefully"
    "Fix Composer Bug (CSS)": [0.05, 0.9]
    "Pinecone Company Docs": [0.2, 0.95]
    "Wire ResponseVerifier": [0.25, 0.8]
    "iOS Font Fix": [0.05, 0.5]
    "Voice Textarea Fix": [0.05, 0.55]
    "Indian Kanoon API": [0.5, 0.85]
    "Legal RAG Build": [0.7, 0.85]
    "file_gen Implementation": [0.4, 0.8]
    "Sarvam STT/TTS": [0.6, 0.7]
    "IPC→BNS Detection": [0.35, 0.9]
    "WhatsApp Integration": [0.85, 0.6]
```

```
    "Hindi OCR (Sarvam Vision)": [0.55, 0.65]
```

10. ANTI-HALLUCINATION ARCHITECTURE

This is the designed (but not yet fully deployed) zero-hallucination pipeline:

```
flowchart LR
    subgraph INPUT["INPUT STAGE"]
        A["User Query"]
        B["Uploaded Document"]
    end

    subgraph STAGE1["STAGE 1: IMAGE PRE-PROCESSING"]
        C["Deskew · Denoise · Binarize<br/>Contrast boost
    end

    subgraph STAGE2["STAGE 2: OCR ENGINE SELECTION"]
        D{"Document type?"}
        D1["PyMuPDF<br/>Digital PDFs<br/>ZERO errors ✓"]
        D2["Sarvam Vision<br/>Hindi/handwritten<br/>95.9%
        D3["PaddleOCR<br/>Mixed Hindi+English<br/>88% acc
        D --> D1 & D2 & D3
    end

    subgraph STAGE3["STAGE 3: OCR POST-PROCESSING"]
        E["Unicode normalize (NFC)<br/>Remove garbage art
    end

    subgraph STAGE4["STAGE 4: RAG RETRIEVAL"]
        F["Translate query to English<br/>Pinecone vector
    end

    subgraph STAGE5["STAGE 5: PROMPT PREPROCESSING"]
        G["Intent classify<br/>Entity extract (sections,
    end
```

```
    subgraph STAGE6["STAGE 6: LLM GENERATION"]
        H["Project Brain<br/>temp=0.1 (documents)<br/>tem
    end

    subgraph STAGE7["STAGE 7: RESPONSE VERIFICATION ✖ NO
        I["Number verification<br/>Response length check<
    end

    subgraph STAGE8["STAGE 8: MULTILINGUAL OUTPUT ✖ NOT
        J["sarvam-translate → user language<br/>Bulbul v2
    end

    A --> STAGE4
    B --> STAGE1 --> STAGE2 --> STAGE3 --> STAGE5
    STAGE4 --> STAGE5 --> STAGE6 --> STAGE7 --> STAGE8

    style D2 fill:#ff4444,color:#fff
    style STAGE7 fill:#ff9900,color:#000
    style STAGE8 fill:#ff9900,color:#000
```

Anti-Hallucination Rulebook

Root Cause	Problem	Fix Required
Knowledge Gap	LLM doesn't know Grizon's own pricing	Populate + wire Pinecone company_docs RAG
Context Confusion	Multiple topics confuse the LLM	XML tags: `<DOCUMENT_CONTENT>`, `<RETRIEVED_KNOWLEDGE>`
OCR Noise	Hindi scans produce garbage text	5-step image preprocessing + Sarvam Vision
Prompt Ambiguity	Vague query → LLM guesses intent	Intent classification + entity extraction

High Temperature	Creativity causes invented facts	temp=0.1 for document Q&A
No Verification	Hallucinations reach user undetected	Wire ResponseVerifier LLM-as-Judge
Legal Citations	AI fabricates plausible case names	Indian Kanoon API mandatory for Legal Agent
IPC→BNS confusion	Wrong section numbers for post-July 2024 cases	Date-based law transition detection

11. CREDIT & BILLING SYSTEM ANALYSIS

Current Credit Flow

```
flowchart TD
    A["User subscribes: Pro ₹1,999/month"] --> B["PhonePe
    B --> C["user_wallets table updated<br/>credits = 1,0

    C --> D{"User sends message"}
    D --> E["BEGIN TRANSACTION<br/>SELECT credits FROM us
    E --> F{"credits ≥ cost?"}
    F -->|"YES"| G["UPDATE user_wallets<br/>SET credits =
    F -->|"NO"| H["Return: Insufficient credits<br/>Show
    G --> I["INSERT INTO messages<br/>tokens_used · credi
    I --> J["COMMIT"]

    subgraph RATES["CONFIRMED CREDIT RATES"]
        R1["Text query: 3 credits · 1.4–3.3s"]
        R2["PDF document: 6 credits · 5.7s"]
    end
```

```
    subgraph LIMITS["RATE LIMITS"]
        L1["60 req/hr"]
        L2["400 req/day"]
        L3["2,000 req/week"]
        L4["10,000 req/month"]
    end
```

> [!WARNING] **UI Confusion: Two credit pools are shown without explanation.** "Plan credits: 6 of 1,000" vs "Credits remaining: 991" — users don't understand the difference. Add tooltip: *"Plan credits reset every month. Wallet credits are your permanent balance."*

---

12. FOUNDER RECOMMENDATIONS

The 5 Most Important Things to Do Right Now

```
graph TD
    P1["#1 PRIORITY — TODAY<br/>Fix Composer Push Bug<br/
    P2["#2 PRIORITY — THIS WEEK<br/>Populate Pinecone + W
    P3["#3 PRIORITY — THIS WEEK<br/>Wire ResponseVerifier
    P4["#4 PRIORITY — LEGAL AGENT<br/>Indian Kanoon API +
    P5["#5 PRIORITY — MONTH 2<br/>Sarvam AI Full Integrat

    P1 --> P2 --> P3 --> P4 --> P5

    style P1 fill:#ff4444,color:#fff
    style P2 fill:#ff6600,color:#fff
    style P3 fill:#ff9900,color:#000
    style P4 fill:#0066cc,color:#fff
    style P5 fill:#006633,color:#fff
```

What Makes Grizon AI's Legal Agent Unique (When Built Right)

Capability	Generic Legal AI	Grizon Legal Agent
Response format	Chat only	Chat summary + Full document report
Analysis depth	Statute lookup	6-stage structured reasoning
Argument balance	One-sided	Both sides argued objectively
India-specific	Generic	BNS/BNSS/BSA + 25 HC jurisdictions
Confidence rating	None	Strong/Moderate/Weak + High/Medium/Low
"What could change everything"	None	Explicitly called out in every analysis
Language	English only	Hindi + 22 Indian languages (via Sarvam)
Scanned documents	Usually fails	Sarvam Vision OCR (when wired)

Market Opportunity

Grizon AI with the Legal Agent + Sarvam integration addresses:

- **250M English speakers** (current) → **1.4 Billion Indians** (with Sarvam)
- **First AI platform** in India with structured 6-stage legal reasoning + voice input in Hindi
- **Addressable legal market**: India has ~60 million cases pending in courts — each party needs accessible legal research

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APPENDIX: KEY TECHNOLOGY

REFERENCES

Technology	Role	Docs
Sarvam Saaras v3	Hindi STT (voice input)	api.sarvam.ai
Sarvam Bulbul v2	Hindi TTS (voice output)	api.sarvam.ai
sarvam-translate v1	22 Indian language translation	api.sarvam.ai
Sarvam Vision	Hindi OCR for scanned docs	api.sarvam.ai
Indian Kanoon API	Free legal case search	indiankanoon.org/api
Pinecone	Vector DB for RAG	pinecone.io
PyMuPDF	Digital PDF extraction	pymupdf.readthedocs.io
PaddleOCR	Mixed Hindi+English OCR	paddlepaddle.org
PhonePe PG	UPI payments (SHA256 X-VERIFY)	developer.phonepe.com
Claude 3.5 Sonnet	Primary LLM (Project Brain)	anthropic.com
DeepSeek V3/ R1	Fallback LLM	api.deepseek.com

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